American Computer Science League

2025 Finals ● Short Problems ● Intermediate/Classroom Division

1. Boolean Algebra

Which ordered triples make the following Boolean expression FALSE?

$$A(\overline{B} + C) + \overline{B}(C + \overline{A}) + \overline{B + C}$$

A. (0,0,1) (0,1,1)
B. (1,1,0) (0,*,1)
C. (1,1,0) (0,1,*)
D. (*,1,0) (0,1,0)
E. None of the above

2. Boolean Algebra

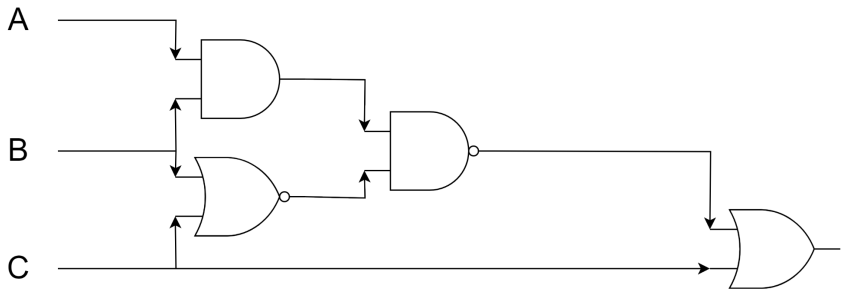
Which of the following is equivalent to this Boolean expression?

$$(\overline{A} + AB)(\overline{B}\,\overline{C}) + (\overline{A} + \overline{BC})$$

A. $\overline{A}\,BC$
B. $A + \overline{BC}$
C. $\overline{A}\,B + C$
D. $\overline{ABC}$
E. None of the above

3. Digital Electronics

What Boolean expression represents the following digital circuit?



A. $\overline{AB(\overline{B} + \overline{C}) + C}$
B. $\overline{(A + B) + \overline{BC}\,C}$
C. $\overline{AB\,(\overline{B + C}) + C}$
D. $\overline{AB}\,(B + C) + \overline{C}$
E. None of the above

4. Bit-String Flicking

Evaluate the following expression:

(NOT 000100) AND (LSHIFT-3 (RCIRC-1 011011))

A. 101000
B. 111011
C. 101101
D. 111001
E. None of the above

5. Bit-String Flicking

How many different values of X, a bit-string of 5 bits, make the following equation TRUE?

(10010 AND (RSHIFT-1 01111)) OR X =
(RCIRC-2 ((LSHIFT-2 01010) OR X))

A. 0
B. 2
C. 4
D. 8
E. None of the above

6. Recursive Functions

Find $f(f(f(20)))$, given:

$$f(x) = \begin{cases} 2 \cdot f(x-3) + 3 & \text{if } x \geq 12 \\ f\left(\left[\frac{x}{2}\right]\right) - 2 & \text{if } 6 \leq x < 12 \\ x \bmod 4 & \text{if } x < 6 \end{cases}$$

Note: $[a]$ is the greatest integer less than or equal to a

A. 13
B. 11
C. 5
D. 1
E. None of the above

7. Recursive Functions

Find $f(20, 5)$, given:

$$f(x,y) = \begin{cases} f\left(\frac{x}{2},\ y+2\right) + 3 & \text{if } x \text{ is even and } y < 8 \\ f\left(\frac{x-1}{2},\ y-3\right) - 1 & \text{if } x \text{ is odd and } y \geq 8 \\ x + y & \text{otherwise} \end{cases}$$

A. 9
B. 14
C. 16
D. 22
E. None of the above

8. Prefix/Infix/Postfix Notation

Convert the following prefix expression to postfix, if all numbers are single digits:

− + a / b c * 3 + * a b c

A. a b c + / 3 a b * c + * −
B. a b c / + 3 a b * c + * −
C. a b c / + 3 a b * c * + −
D. a b c / + 3 a b + c * * −
E. None of the above

9. Prefix/Infix/Postfix Notation

Evaluate the following prefix expression, if all numbers are single digits:

+ / * 2 + 6 4 − 9 / 8 2 ^ 4 2

A. 4
B. 10
C. 16
D. 20
E. None of the above

10. Computer Number Systems

How many 4s appear in the octal numbers from 32_8 to 64_8, inclusive?

A. 10
B. 11
C. 12
D. 13
E. None of the above

11. Computer Number Systems

What is the rightmost octal digit of 3_8 ^ 7_{10}?

A. 0
B. 1
C. 3
D. 5
E. None of the above

12. Data Structures

Build a binary search tree for:

PRESIDENT

After deleting the root of the tree, how many nodes have exactly one child?

A. 2
B. 3
C. 4
D. 5
E. None of the above

13. Data Structures

Given the string "WATERMELON", perform the following algorithm on an initially empty stack.

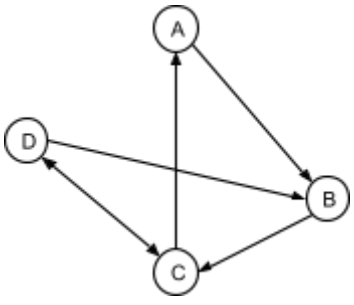
```
for each letter in the word
        if the letter is a vowel
                x = pop()
        else
                push(letter)
next letter
```

What does the stack look like from top to bottom after the algorithm is finished?

A. N
B. R
C. R
   N
D. N
   R
E. None of the above

14. Graph Theory

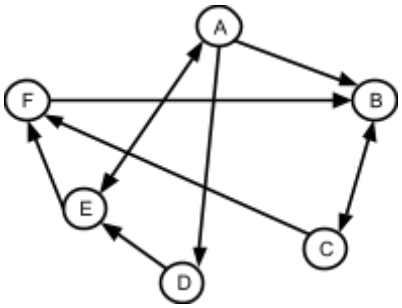
Given the directed graph below, how many paths of length 3 are there?



A. 10
B. 12
C. 14
D. 16
E. None of the above

15. Graph Theory

How many cycles are there in the following directed graph?



A. 4
B. 5
C. 6
D. 7
E. None of the above

16. What Does This Program Do?

What is the output of the following program?

```
n = 2025
p = 1
s = 0
while n != 0
     d = n % 10
     s = s + d * p
     p = p * 8
     n = int(n / 10)
end while
output s
```

A. 549
B. 1045
C. 1301
D. 3751
E. None of the above

17. LISP

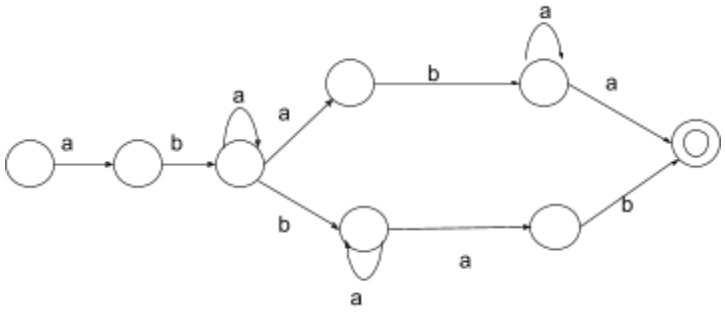
Evaluate the following LISP expression:

(CADAR (REVERSE (CDDR '((a b) c (b (a c)) (a (b) (c))))))

A. b
B. (b)
C. ((b) (c))
D. (a c)
E. None of the above

18. FSAs and Regular Expressions

How many of these strings are accepted by the following FSA?



- abaaba
- abbaaba
- ababa
- abbabb
- ababaaa
- abbaaa

A. 0
B. 1
C. 2
D. 3
E. None of the above

19. FSAs and Regular Expressions

The soup of the day is vegetable soup. Only vegetables which are accepted by the following regular expression can be in the soup. How many vegetables can be used?

[^bs] [aeiou]* [bpr]* [aeiou]* [e-t]* s?

onion	carrot	turnip	peas
asparagus	celery	spinach	beans
broccoli	beet	cauliflower	cabbage

A. 4
B. 5
C. 6
D. 7
E. None of the above

20. Assembly Language Programming

What is the output when the following program is executed?

```
T         DC        40
N         DC        2025
TOP       LOAD      T
          ADD       =1
          STORE     T
          LOAD      N
          DIV       T
          MULT      T
          SUB       N
          BE        STOP
          BU        TOP
STOP      PRINT     T
          END
```

A. 41
B. 43
C. 45
D. 47
E. None of the above